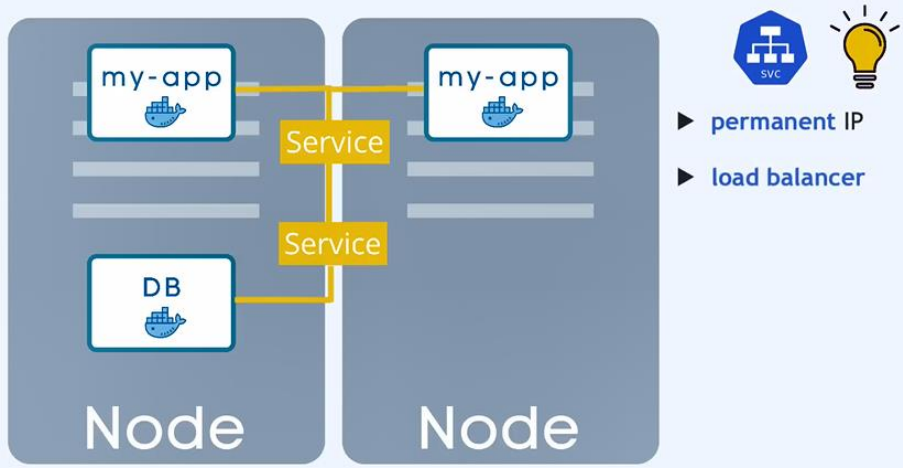
The ***advantage of using distributed systems*** or ***microservices*** with ***containers*** is that we have many replicas of our app, so if one of them is crashed or down, the others ***will still handle the requests of users***

****

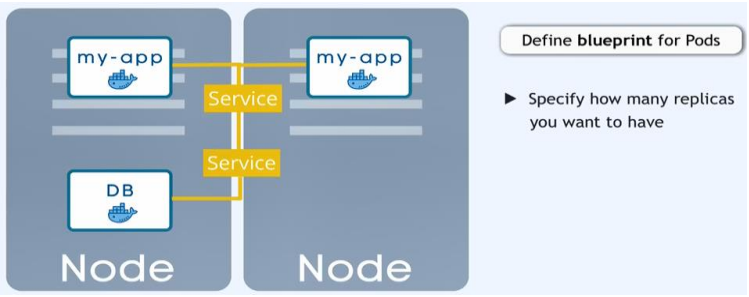
The other advantage of ***service*** that it ***contains load balancer***

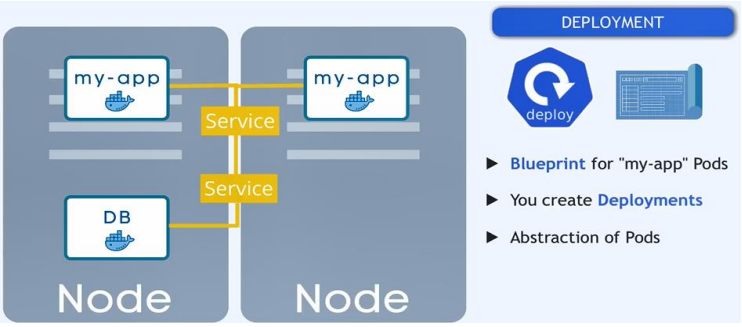
****



****

This blueprint called deployment, they are abstractions of Pod



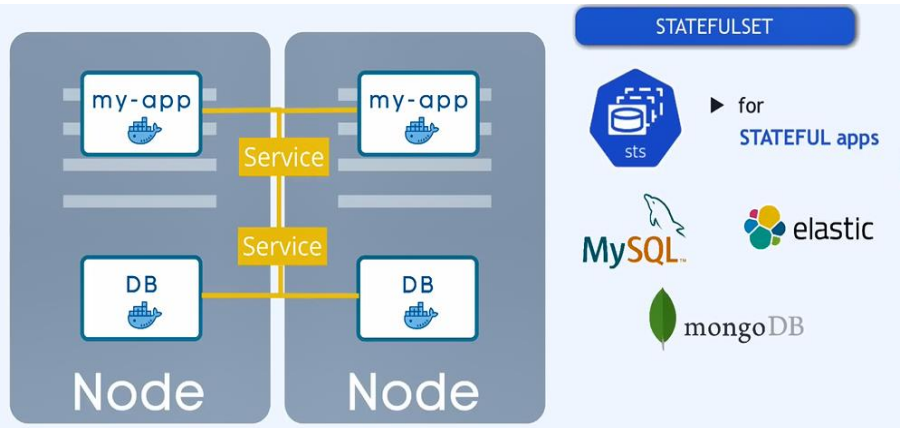


****

We can't replicate the DB using deployments, because it has its state (data with volume):

****

The solution for previous is:

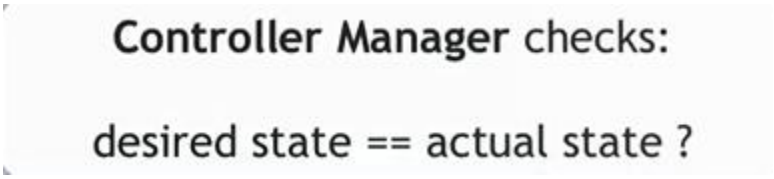


****

But the deployment for DB in same cases may not be easy

****

If one of the replicas die, then:



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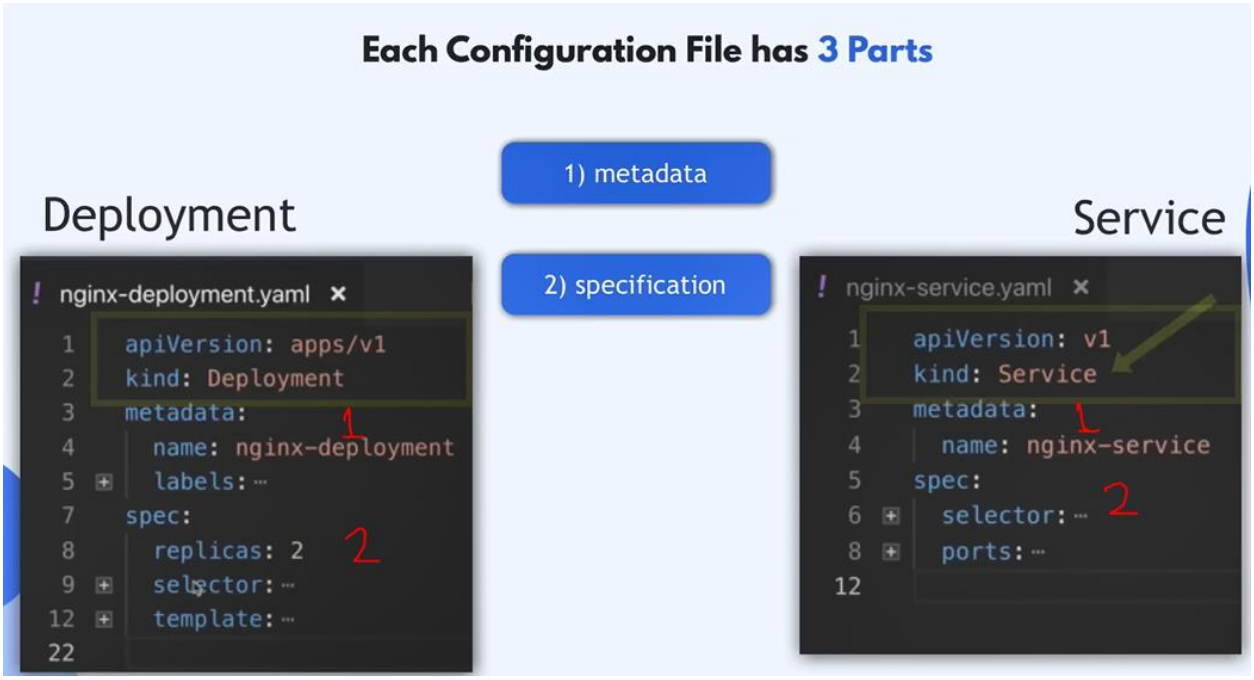

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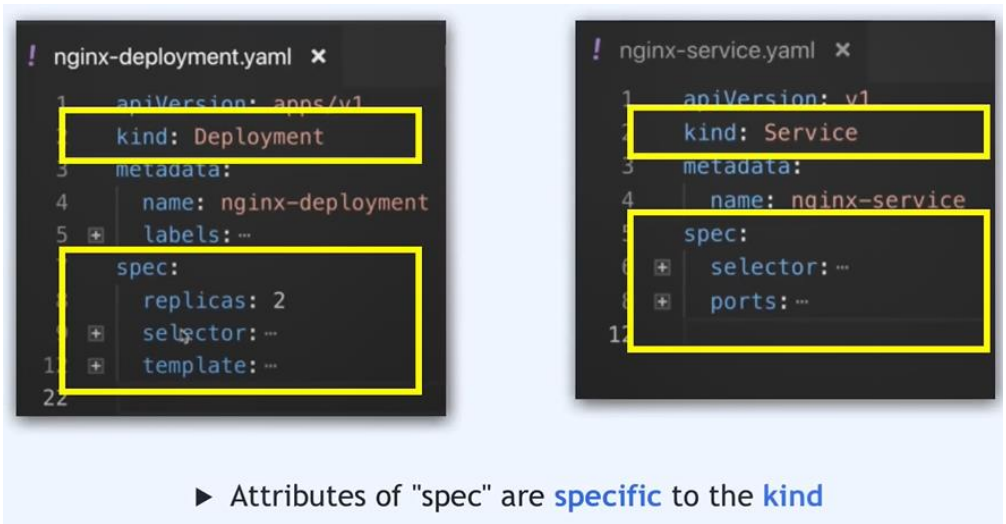
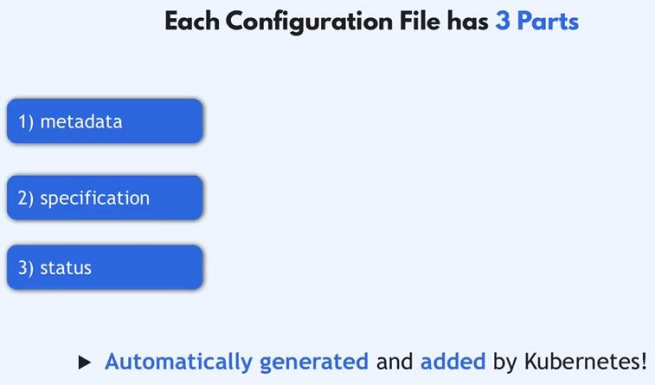


****



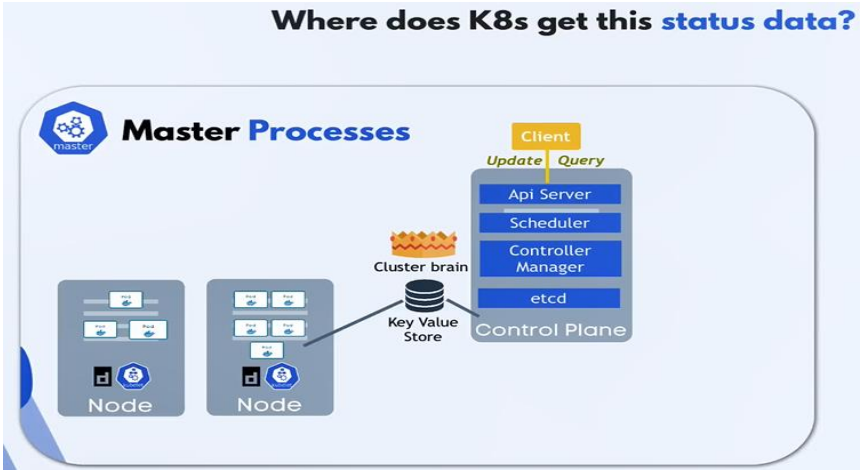
****

For status, kubernetes will always check the desired && actual status:

- Then we have choice of the desired = actual then it will still works
- Else, kubernetes will try to solve the problem.

****

Kubernetes save the important data (like status) in etcd (key/value store)



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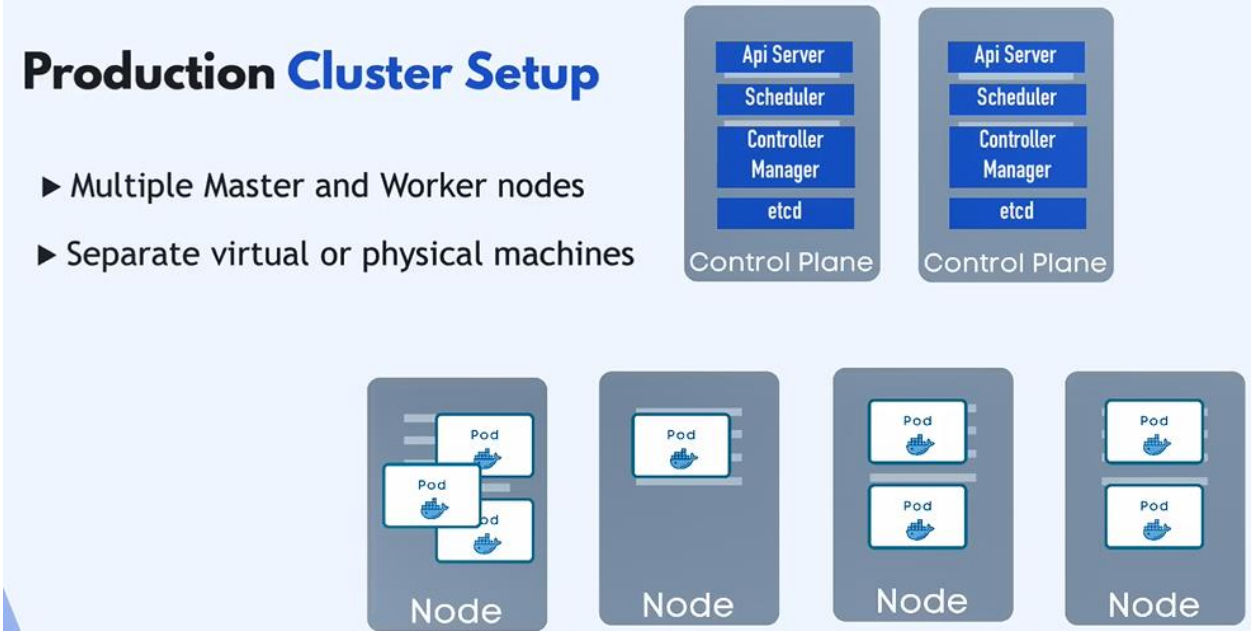
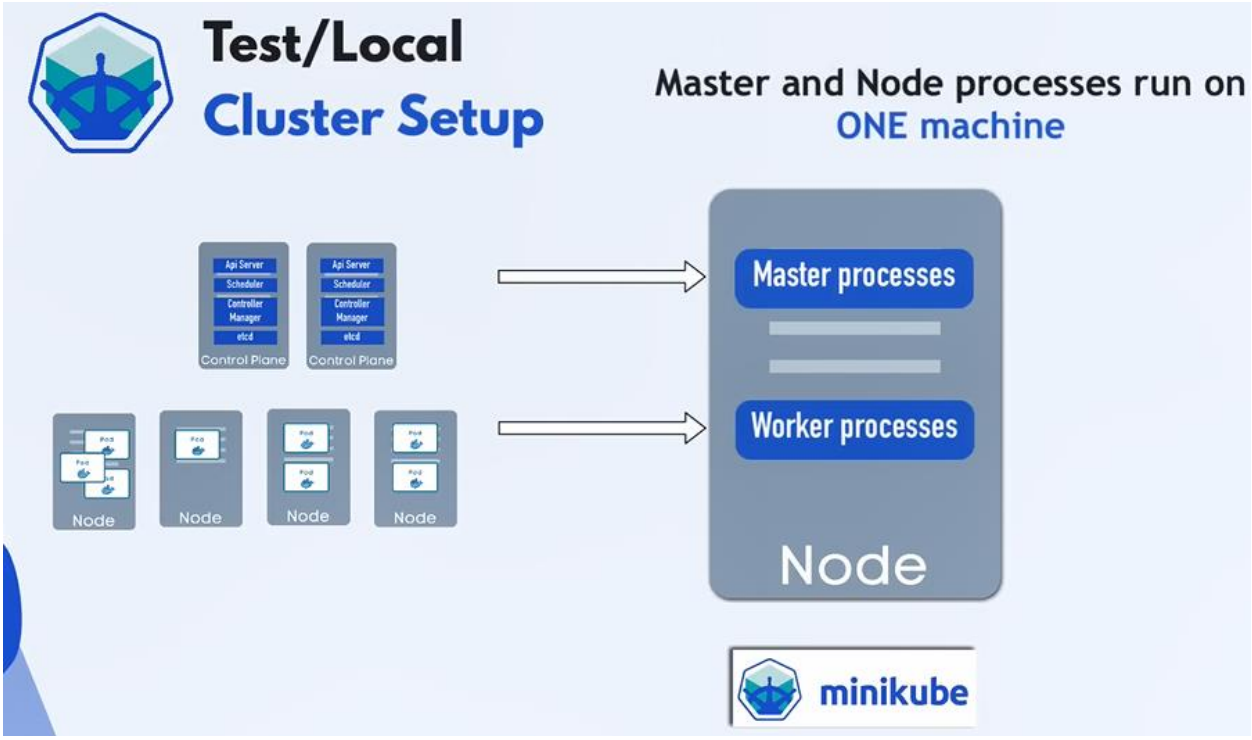


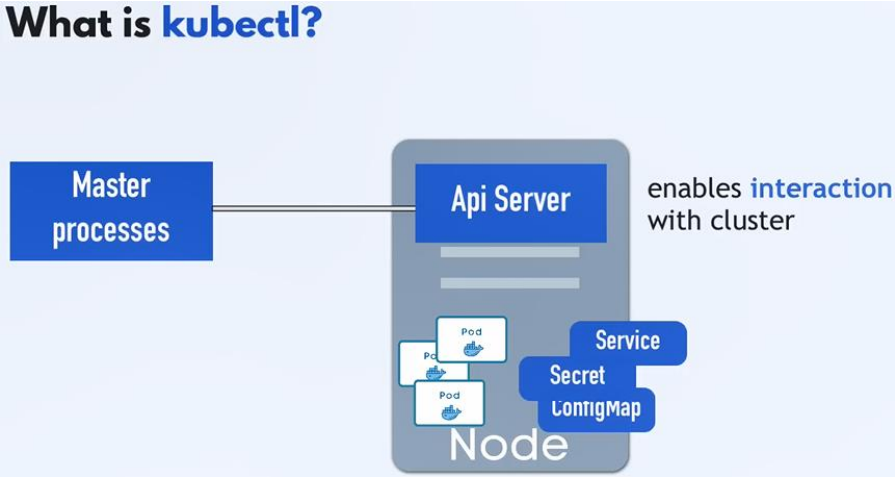
****

In test environment the ***master*** && ***worker processes*** are running ***on the same node***:



****

The main entry for kubernetes cluster is Api Server:



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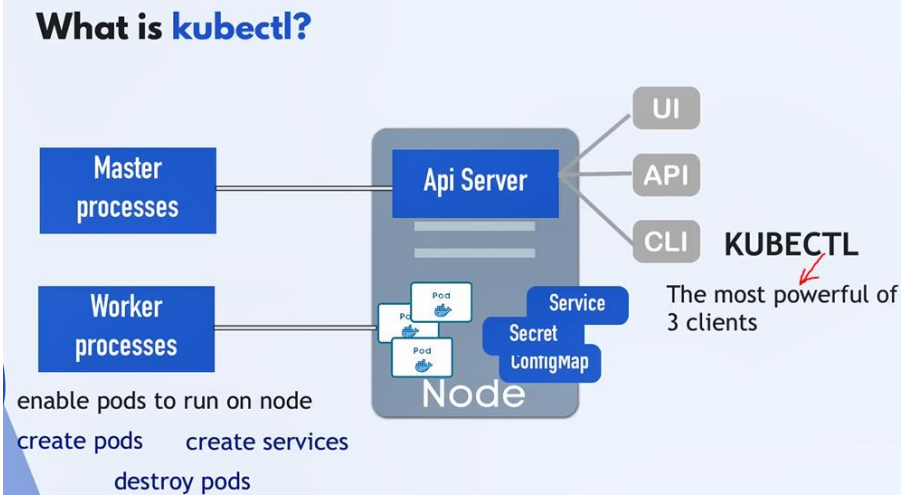

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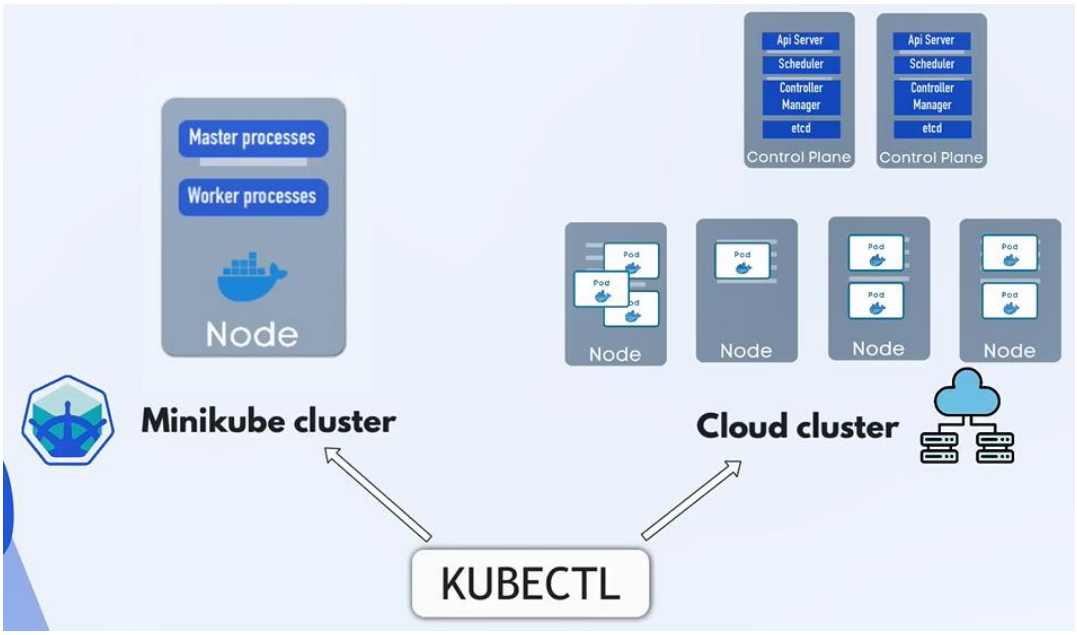


****

Deployment Configuration File

- kind: "Deployment"

Main Part:
Blueprint for Pods

- template: configuration for Pod

Why?
Because Deployment manages Pod

****

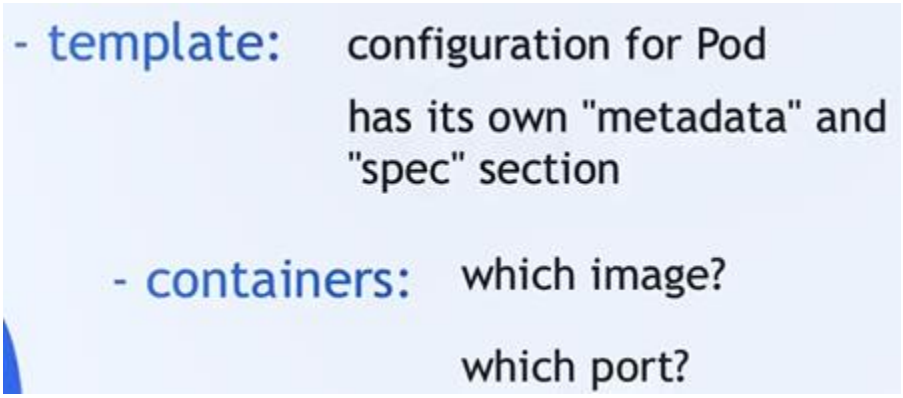
Main Part:
Blueprint for Pods

- template: configuration for Pod

has its own "metadata" and "spec" section

****

For spec in template we may have multiple containers per pod, but we usually have one main app per pod:



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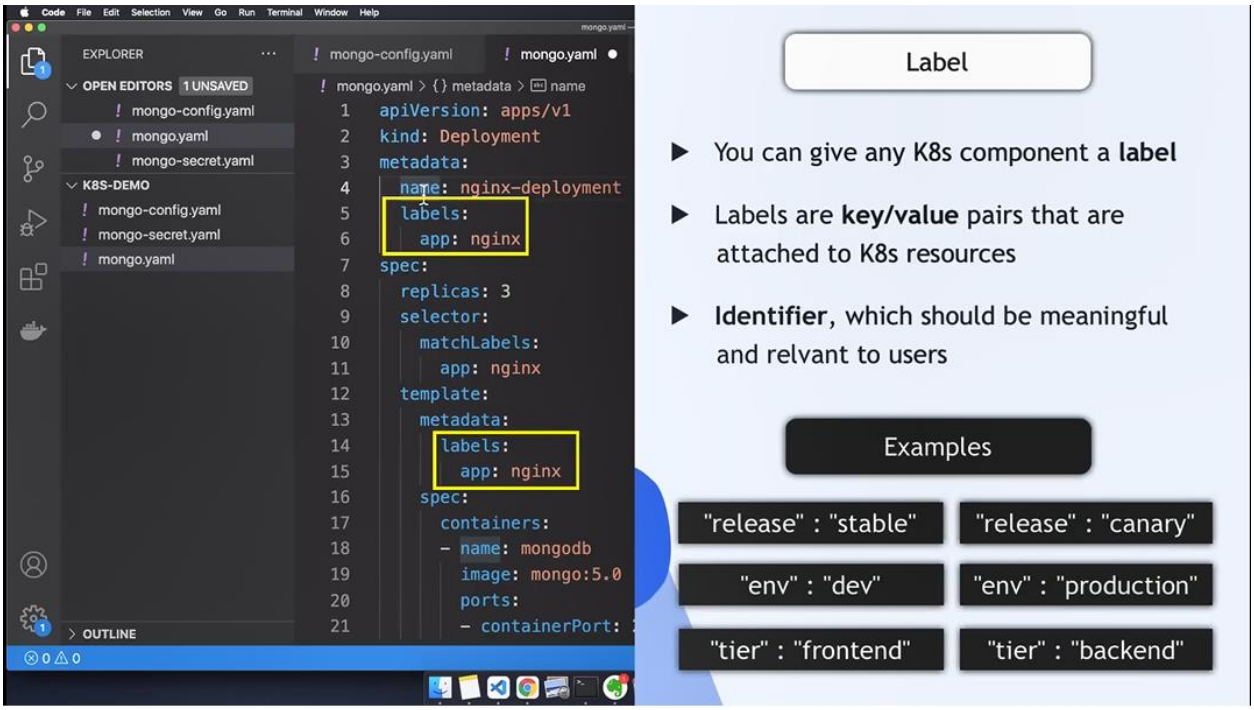

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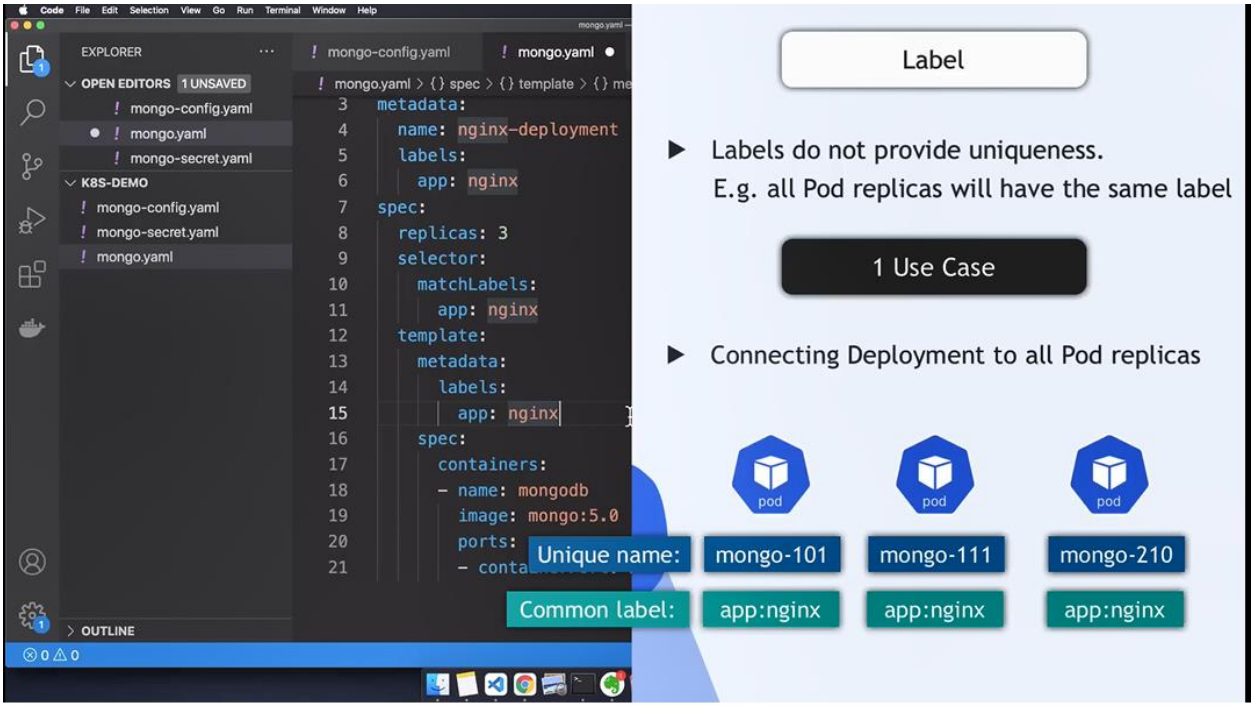
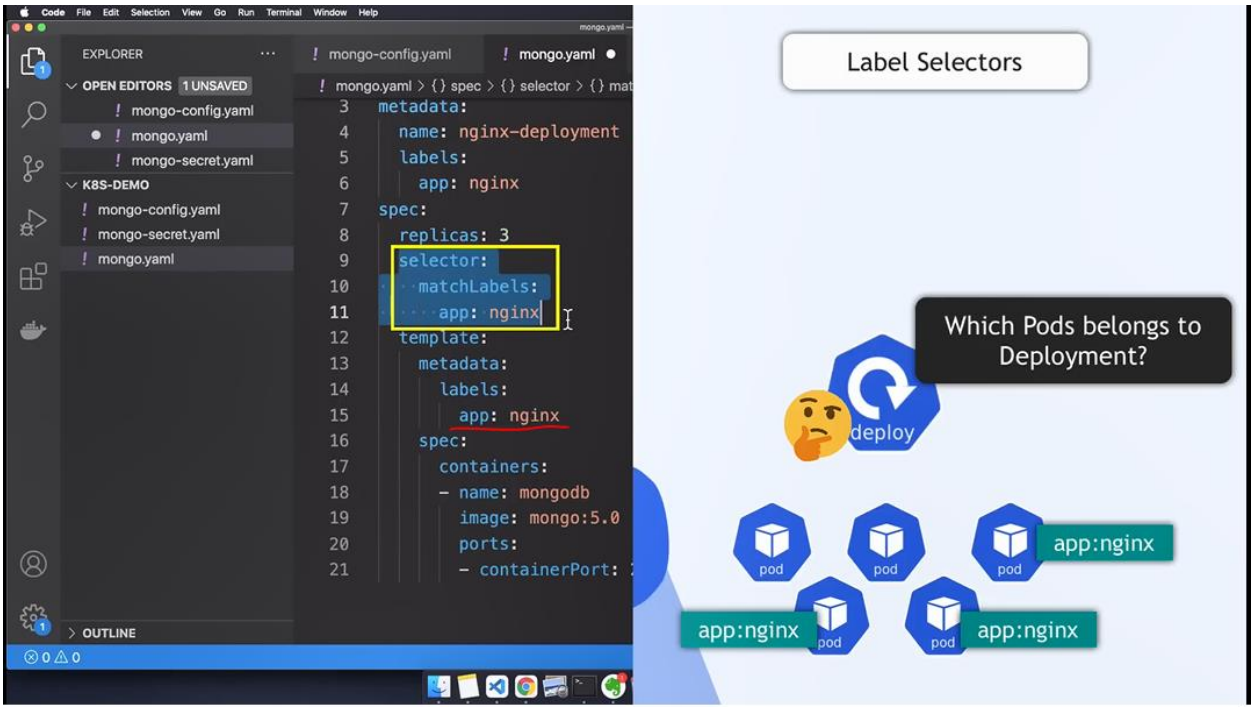
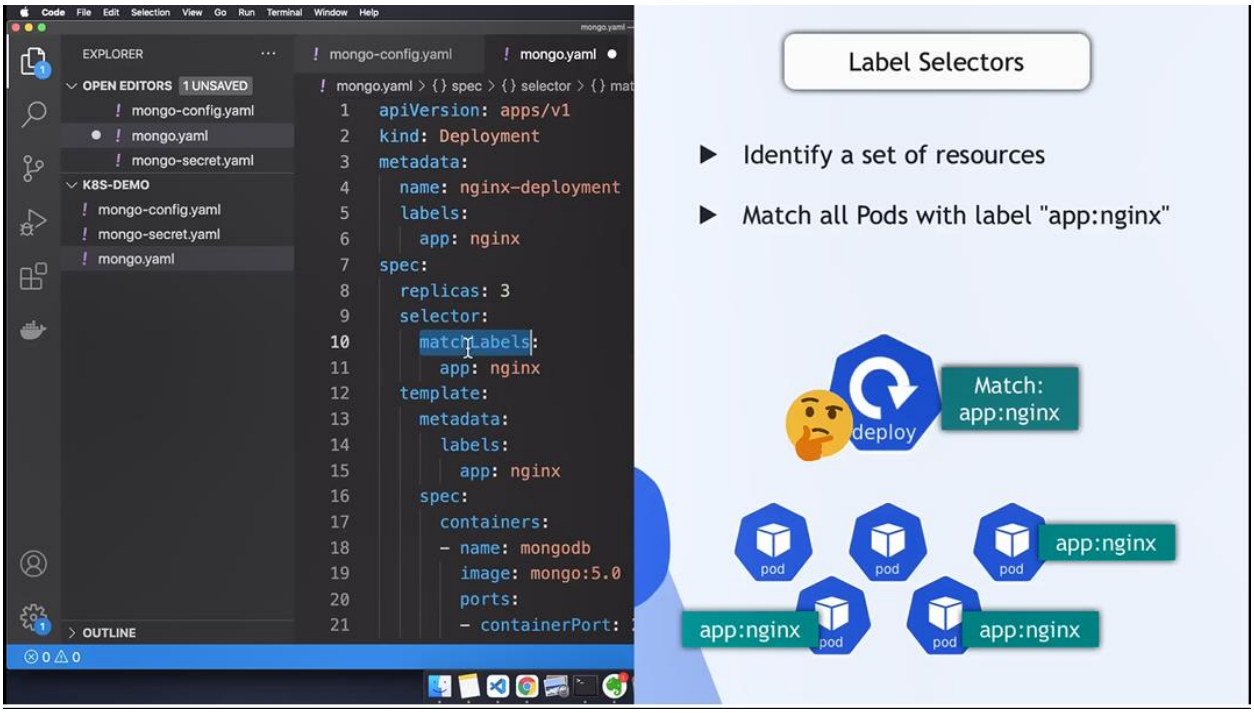

Label
Labels do not provide uniqueness.
E.g. all Pod replicas will have the same label
1 Use Case
Connecting Deployment to all Pod replicas
pod
Unique name:
mongo-101
mongo-111
mongo-210
Common label:
app:nginx
app:nginx
app:nginx
EXPLORER
OPEN EDITORS 1 UNSAVED
mongo-config.yaml
mongo.yaml
mongo-secret.yaml
K8S-DEMO
OUTLINE

****


Label Selectors
Which Pods belongs to Deployment?
deploy
pod
app:nginx
app:nginx
app:nginx
EXPLORER
OPEN EDITORS 1 UNSAVED
mongo-config.yaml
mongo.yaml
mongo-secret.yaml
K8S-DEMO
OUTLINE

****


Code File Edit Selection View Go Run Terminal Window Help
EXPLORER
OPEN EDITORS 1 UNSAVED
mongo-config.yaml
mongo.yaml
mongo-secret.yaml
K8S-DEMO
mongo-config.yaml
mongo-secret.yaml
mongo.yaml
OUTLINE
mongo.yaml > { } spec > { } selector > { } mat
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4 name: nginx-deployment
5 labels:
6 app: nginx
7 spec:
8 replicas: 3
9 selector:
10 matchLabels:
11 app: nginx
12 template:
13 metadata:
14 labels:
15 app: nginx
16 spec:
17 containers:
18 - name: mongodb
19 image: mongo:5.0
20 ports:
21 - containerPort:
Label Selectors
Identify a set of resources
Match all Pods with label "app:nginx"
deploy
Match:
app:nginx
pod
app:nginx
app:nginx
app:nginx